

JAYANTHAN SENTHILKUMAR

Software Engineer | Backend Developer | AI & Machine Learning Engineer

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Current Location: **India** (open to relocation to Australia) | **Visa Status:** Seeking employer-sponsored visa

Target Market: Australia - open to relocation and remote/hybrid arrangements

PROFESSIONAL SUMMARY

Entry-level Software Engineer based in India, seeking to relocate to Australia to build a long-term career in software and AI engineering. Delivered distributed backend systems, RESTful APIs, and full-stack applications using PHP, Laravel, Python, and MySQL, including a 30% reduction in backend latency and a 40% improvement in API reliability during an industrial trainee placement. Built three end-to-end projects spanning AI-assisted trip optimisation, IoT-enabled fleet management, and enterprise ERP systems. Completed a Bachelor of Technology in Artificial Intelligence and Machine Learning (CGPA 8.08) while building applied machine learning and cloud deployment capability. Brings strong engineering fundamentals and a genuine commitment to contributing to Australia's technology sector as an international graduate hire.

CORE COMPETENCIES

Backend Development • Full-Stack Web Development • RESTful API Design • System Design • Database Design & Optimisation • Problem Solving • Analytical & Critical Thinking • Technical Communication • Cross Functional Collaboration • Agile-Aligned Delivery • Continuous Learning

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, PHP, HTML, CSS, Java

Backend Technologies: MVC Architecture, Object-Oriented Programming, JWT Authentication, WebSockets

Frameworks: Laravel, Flask

Databases: MySQL (query optimisation, indexing, schema design), SQLite

Enterprise Technologies: SAP ABAP Cloud, SAP BTP, ABAP RAP, SAP HANA Cloud

Software Engineering: SDLC, Agile Methodologies, API Integration, Technical Documentation, Authentication & Authorization, JSON

AI & Machine Learning: Predictive Modelling, Applied ML-Assisted Feature Design, Data-Driven Decision Logic (project-level application)

Developer Tools & Version Control: Git, GitHub, GitHub Actions (CI/CD)

Other: IoT integration (Arduino), cross-functional collaboration

PROFESSIONAL EXPERIENCE

Technology Innovation Hub, MKCE

April 2024 - July 2025

Software Developer Trainee

- Engineered a distributed backend system using PHP and SQL to support a growing user base, reducing service latency by 30% and increasing overall system throughput.
- Designed and developed RESTful APIs to standardise inter-service communication, improving system reliability by 40% and reducing integration errors across connected modules.
- Optimised MySQL queries and indexing strategy to resolve a data processing bottleneck, cutting processing time by 35% and improving application responsiveness under load.

EDUCATION

Bachelor of Technology, Artificial Intelligence and Machine Learning

2022 – 2026

M. Kumarasamy College of Engineering, Karur, India | CGPA: 8.08/10

PROJECTS

PLANGO - AI Assisted Trip Planner

Tech Stack: Flask, Python, SQLite, JavaScript | GitHub: github.com/jayanthansenthilkumar/PlanGo

- Built a trip-optimisation web application that ingests real-time traffic data to generate route recommendations, improving route accuracy by 40% and reducing computed route latency by 60%.
- Designed predictive logic for traffic and travel-time estimation, contributing to a 35% increase in measured user satisfaction during testing.

SPARK - An Event Management Platform

Tech Stack: PHP, MySQL, JavaScript, HTML, CSS | GitHub: github.com/jayanthansenthilkumar/Spark

- Improved event management efficiency by 50% by architecting a role-based platform with modules for registrations, project submissions, evaluation workflows, scheduling, and approvals.
- Reduced administrative effort by 40% by developing centralized dashboards, automated announcements, project tracking, and MySQL-backed data management for students, coordinators, judges, and administrators.

Ishaa - Python based on AI Framework

Tech Stack: Python, ASGI, SQLite, WebSockets, GraphQL, OpenAPI, JWT Authentication, REST APIs, CLI, GitHub Actions | GitHub: github.com/jayanthansenthilkumar/Ishaa

- Reduced web application development effort by 60% by architecting a zero-dependency ASGI-native Python framework featuring a custom ORM, JWT authentication, WebSocket support, GraphQL, dependency injection, background tasks, OpenAPI documentation, and an integrated CLI.
- Improved application performance and deployment reliability by 45% by engineering proprietary SARE (Self-Evolving Adaptive Routing Engine), RMF (Reality-Mode Framework), and SEQP (Self-Evolving Quality Pipeline) to enable intelligent routing optimization, multi-reality execution, automated risk analysis, test generation, and CI/CD pipeline evolution.

Cargo Fleet Management System

Tech Stack: Laravel, Python, Arduino, IoT Sensors

- Architected an IoT-integrated fleet management system combining sensor data pipelines with load-prediction logic, improving fleet utilisation by 55%.
- Implemented real-time vehicle tracking and safety-monitoring pipelines, improving transport reliability and reducing logistics delays by 35%.

KR Connect - Smart College ERP

Tech Stack: PHP, MySQL, Ajax, jQuery, HTML, CSS, JavaScript | Live: krconnect.mkce.ac.in

- Developed Discipline and Grievance Management modules using PHP, Ajax, and MySQL, streamlining issue tracking and record updates by 50% for the college administration.
- Built real-time administrative dashboards using HTML, CSS, and JavaScript, improving page load speed and operational efficiency by 45%.

CERTIFICATIONS

- GitHub Foundations - GitHub
- SAP - ABAP Backend Development - SAP
- Google Play & Unity - Associate Game Development - Unity Technologies

LEADERSHIP & ACHIEVEMENTS

- Finalist, PECHACKS 2.0 Hackathon
- Guest Lecturer, "Git vs GitHub" Workshop - Panimalar Engineering College, Chennai.
- Recipient, Student of the Month - Technical Leadership & Performance